

variable gives the fraction of the total population in the grid cell for a given race-ethnic group that comes from the “source” county.

When a grid cell lies completely within a county, then the fraction will be 1. When a grid cell is in more than county, then the sum of the fractions across the counties for a given race-ethnic group must sum to one. In Table J-5, you can see that for grid cell (TargetCol=123, TargetRow=18) that the fraction of Asian Non-Hispanic coming from county (SourceCol=16, SourceRow=71) is 0.49 and for county (SourceCol=49, SourceRow=3) the fraction is 0.51. In this case, about half the population of Asian Non-Hispanics comes from each of the two counties. In the case of Black Hispanics, the fraction from county (SourceCol=16, SourceRow=71) is only 0.12, with most Black Hispanics in this grid cell coming from county (SourceCol=49, SourceRow=3).

Table J-5. Population-Weight File Fragment from PopGrid

SourceCol	SourceRow	TargetCol	TargetRow	Race	Ethnicity	Value	Year
16	71	123	18	ASIAN	NON-HISPANIC	0.49	2000
16	71	123	18	ASIAN	HISPANIC	0.21	2000
16	71	123	18	BLACK	NON-HISPANIC	0.49	2000
16	71	123	18	BLACK	HISPANIC	0.12	2000
16	71	123	18	NATAMER	NON-HISPANIC	0.98	2000
16	71	123	18	NATAMER	HISPANIC	0.43	2000
16	71	123	18	WHITE	NON-HISPANIC	0.23	2000
16	71	123	18	WHITE	HISPANIC	0.06	2000
49	3	123	18	ASIAN	NON-HISPANIC	0.51	2000
49	3	123	18	ASIAN	HISPANIC	0.79	2000
49	3	123	18	BLACK	NON-HISPANIC	0.51	2000
49	3	123	18	BLACK	HISPANIC	0.88	2000
49	3	123	18	NATAMER	NON-HISPANIC	0.02	2000
49	3	123	18	NATAMER	HISPANIC	0.57	2000
49	3	123	18	WHITE	NON-HISPANIC	0.77	2000
49	3	123	18	WHITE	HISPANIC	0.94	2000
6	23	45	1	ASIAN	NON-HISPANIC	0.00	2000
6	23	45	1	ASIAN	HISPANIC	0.00	2000
6	23	45	1	BLACK	NON-HISPANIC	0.00	2000
6	23	45	1	BLACK	HISPANIC	0.00	2000

SourceCol	SourceRow	TargetCol	TargetRow	Race	Ethnicity	Value	Year
6	23	45	1	NATAMER	NON-HISPANIC	1.00	2000
6	23	45	1	NATAMER	HISPANIC	1.00	2000
6	23	45	1	WHITE	NON-HISPANIC	1.00	2000
6	23	45	1	WHITE	HISPANIC	1.00	2000
6	23	45	2	ASIAN	NON-HISPANIC	1.00	2000
6	23	45	2	ASIAN	HISPANIC	1.00	2000
6	23	45	2	BLACK	NON-HISPANIC	1.00	2000
6	23	45	2	BLACK	HISPANIC	1.00	2000
6	23	45	2	NATAMER	NON-HISPANIC	1.00	2000
6	23	45	2	NATAMER	HISPANIC	1.00	2000
6	23	45	2	WHITE	NON-HISPANIC	1.00	2000
6	23	45	2	WHITE	HISPANIC	1.00	2000

J.2 Monitor Data in U.S. Setup

BenMAP-ready data files were created from 2000 through 2018, as reported to the U.S. Environmental Protection Agency's (EPA) Air Quality System (AQS) for PM_{2.5} and ozone. Table J-6 summarizes the data sources and vintage of the processed data.

Table J-6. Underlying Data Sources for BenMAP Air Quality Data File

Pollutant	AQS Parameter Code	Year	Data Source
PM _{2.5}	88101	2000-2018	https://www.epa.gov/aqs
Ozone	44201	2000-2018	

J.2.1 Data Processing

IEc developed an R script to process the raw AQS data into formatted BenMAP-CE monitor files. The R script completes three major tasks:

1. Padding the raw data to generate a full, daily time series for each pollutant (daily 24 hour average values for PM_{2.5} and daily 8-hour maximum average values for ozone),
2. Converting the data from a long to a wide format, and
3. Adjusting and reformatting the data fields to match BenMAP-CE's expected format for monitor datasets.

For daily data, BenMAP-CE expects the monitor data input file to contain a comma-separated string of 365 air quality values—one value for each day in the calendar year. Within this string, any missing values are represented by a period (“.”). In contrast, the AQS data only contain records for dates with measured air quality values or calculated pollutant metrics (24 hour average concentrations of PM_{2.5} or 8-hour maximum average concentrations for ozone). The raw AQS data are padded so that there is ultimately a row for every day in the year for each monitor, with periods inserted by the R code for every missing value. During this process, the R script also cleans up the raw data by removing any monitors located in Canada, removing any daily values with an Observation Percent less than 75%, and limiting values in the “Event Type” field to “None”.⁴⁴ In addition, PM_{2.5} data are limited to values with a “Sample Duration” of “24 HOUR” or “24-HR BLK AVG”.⁴⁵

Once the dataset has been expanded to the correct volume, the R script then casts it from a long format to a wide format. This means that the R script takes the 365 rows associated with a monitor and converts them to a single comma-separated string needed for the BenMAP-CE “Values” field.

- Data are reported in µg/m³ for PM_{2.5}, and ppb for ozone.

J.2.2 Output Files

Table J-7 lists the number of monitors by pollutant and year, represented in the resulting BenMAP-ready data files. The AQS Parameter Code for PM_{2.5} is 88101 and for Ozone is 44201.

Table J-7. Number of Monitors by Pollutant and Year Included in BenMAP

Year	Number of Monitors by Year	
	PM _{2.5}	Ozone
2000	1,316	1,144
2001	1,355	1,190
2002	1,397	1,195
2003	1,421	1,213
2004	1,253	1,208
2005	1,285	1,198

⁴⁴ The Observation Percent field describes the volume of hourly data used to calculate each daily metric in the pre-generated AQS data files. Based on guidance provided by AQS staff, IEC limited values to those calculated with at least 75% of the underlying measurements. The “Event Type” field identifies days that may have been impacted by exceptional events, such as wildfires. Limiting the data to “Event Type” values of “None” limits the data included in BenMAP-CE to those days representing ambient air quality.

⁴⁵ For PM_{2.5} these values are taken from the “Arithmetic Mean” field of raw AQS data, while for O₃ they are taken from the “1st Max” field, based on guidance from AQS staff. The “1st Max” field is converted from ppm to ppb; BenMAP uses ppb as the unit for ozone concentrations.

Year	Number of Monitors by Year	
	PM _{2.5}	Ozone
2006	1,230	1,205
2007	1,191	1,212
2008	1,196	1,215
2009	1,273	1,214
2010	1,227	1,228
2011	1,137	1,283
2012	1,158	1,285
2013	1,190	1,293
2014	1,315	1,287
2015	1,373	1,276
2016	1,559	1,292
2017	1,463	1,287
2018	1,494	1,275

J.3 Demographic Datasets in U.S. Setup

The U.S. setup includes county-level data on educational attainment, poverty status, health insurance coverage, occupational status, and linguistic isolation. We describe the data sources and processing methodology for each dataset below. All estimates were generated at the county level for 3,108 counties in the contiguous United States.⁴⁶

J.3.1 Educational Attainment

We use data from the ACS to provide county-level summaries of educational attainment. These data represent 5-year average ACS estimates from 2018 to 2022. Specifically, the data included in BenMAP-CE span two broad education categories: no high school diploma (termed “no_hs_degree”), and high school diploma (or equivalency) and above (termed “hs_degree_plus”). The latter category includes individuals with a high school diploma (or equivalency), some college, college degree, or post-graduate degree and ranges from 53-98%.

⁴⁶In 2022, the State of Connecticut changed their county-equivalents from eight counties to nine planning regions (known as Councils of Governments), which have substantial boundary differences. By 2024, all Census Bureau products use the new county-equivalent boundaries, except for 2020 Decennial Census data publications published before June 1, 2022. Due to BenMAP-CE’s grid definition, we continue to include the eight Connecticut counties in this update. We convert any Connecticut planning region data to the original county-equivalent by applying a population weighted adjustment. For further information, please see the Census website: <https://www.census.gov/programs-surveys/geography/technical-documentation/county-changes.html>